



CSIC
2026

GUSTAVO.LINAN@CSIC.ES

IMSE
-cnm



Instituto de
Microelectrónica
de Sevilla

Módulo 3: Trabajando con Datos



OBJETIVOS:

1. Manipular y analizar datos utilizando **Pandas**.
2. Visualizar datos con **Matplotlib**.
3. Aplicarlo a ejemplos de ciencia de datos.





 pandas

About us ▾Getting started

pandas

pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language.

[Install pandas now!](#)

Getting started

- [Install pandas](#)
- [Getting started](#)
- [Try pandas online](#)

Documentation

- [User guide](#)
- [API reference](#)
- [Contributing to pandas](#)
- [Release notes](#)

Community

- [About pandas](#)
- [Ask a question](#)
- [Ecosystem](#)

With the support of:

The full list of companies supporting *pandas* is available in the [sponsors page](#).

```
@software{
  reback2020pandas, author = {The pandas development team},
  title = {pandas-dev/pandas: Pandas},
  month = feb, year = 2020,
  publisher = {Zenodo},
  version = {latest},
  doi = {10.5281/zenodo.3509134},
  url = {https://doi.org/10.5281/zenodo.3509134}
}
```

¿Pandas?

Es una **librería de Python especializada en análisis y manipulación de datos**. Su nombre viene de *Panel Data*, un término usado en estadística y econometría.

Con Pandas podremos:

- **Leer datos** desde muchos formatos (CSV, Excel, bases de datos, etc.).
- **Guardar datos** procesados en archivos para compartirlos.
- **Limpiar y transformar** la información (por ejemplo, eliminar valores faltantes o convertir texto en números).
- **Hacer cálculos rápidos** (medias, máximos, mínimos, sumas, correlaciones...).
- **Seleccionar** solo las partes de los datos que nos interesan.
- **Agrupar y resumir** información (ej. calcular la media de temperaturas por año).
- **Integrarse con librerías de visualización** como **Matplotlib** para generar gráficos.

<https://pandas.pydata.org>

- Nos permite **trabajar con datos como si fueran tablas de Excel**, pero con la potencia de la programación en Python.
- Es **rápido y eficiente**, porque está construido sobre otra librería llamada **NumPy**, optimizada para cálculos numéricos.
- Es **ampliamente usado en ciencia, ingeniería y datos abiertos**, lo que facilita compartir y reproducir el trabajo.

https://pandas.pydata.org/pandas-docs/stable/getting_started/index.html#getting-started

<https://github.com/pandas-dev/pandas/tree/main>

cheat sheets: (chuletas)

- Una cheat sheet (hoja de referencia rápida) **es un resumen visual** que concentra los comandos, funciones o conceptos más importantes de una herramienta o librería.
- **No reemplaza a la documentación** completa ni a la práctica, pero resulta muy útil como apoyo inmediato cuando estamos trabajando y necesitamos recordar la sintaxis o el uso básico de una función.
- Hoy en día, además de estas hojas de referencia, también es fundamental **familiarizarnos con el uso de modelos de lenguaje** como ChatGPT, Gemini o Microsoft Copilot. Estas herramientas pueden actuar como referencias rápidas y contextuales, ayudándonos a resolver dudas concretas de manera eficiente.



with pandas Cheat Sheet
<http://pandas.pydata.org>

Creating DataFrames

	a	b	c
1	4	7	10
2	5	8	11
3	6	9	12

```
df = pd.DataFrame(  
    [[4, 7, 10],  
     [5, 8, 11],  
     [6, 9, 12]],  
    index=[1, 2, 3],  
    columns=['a', 'b', 'c'])
```

Specify values for each row.

		a	b	c
N	v			
D	1	4	7	10
	2	5	8	11
e	2	6	9	12

Create DataFrame with a MultiIndex

```
df = (pd.melt(df)
```

```

.rename(columns={
    'variable': 'var',
    'value': 'val'})
.query('val >= 200') )

```

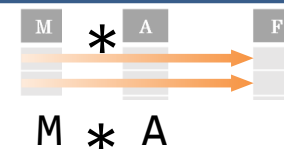
F	M	A

Each **variable** is saved
in its own **column**

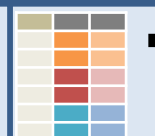


Each **observation** is
saved in its own **row**

Tidy data complements pandas's **vectorized operations**. pandas will automatically preserve observations as you manipulate variables. No other format works as intuitively with pandas.



`pd.concat([df1,df2])`
Append rows of DataFrames



`pd.concat([df1,df2], axis=1)`
Append columns of DataFrames

- df.[head](#)(n)
Select first n rows.
- df.[tail](#)(n)
Select last n rows.


```
df[['width', 'length',  
   'species']]  
    Select multiple columns with specific  
    names.  
df['width'] or df.width  
    Select single column with specific name.  
df.filter(regex='regex')
```

```
df.query('Length > 7')
df.query('Length > 7 and Width < 8')
df.query('Name.str.startswith("abc")')
```

Use `df.loc[]` and `df.iloc[]` to select only rows, only columns or both.
Use `df.at[]` and `df.iat[]` to access a single value by row and column.
First index selects rows, second index columns.

```
df.iloc[10:20]
    Select rows 10-20.

df.iloc[:, [1, 2, 5]]
    Select columns in positions 1, 2 and 5 (first
    column is 0).

df.loc[:, 'x2':'x4']
    Select all columns between x2 and x4
    (inclusive).

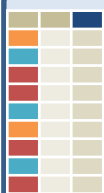
df.loc[df['a'] > 10, ['a', 'c']]
    Select rows meeting logical condition, and only
    the specific columns .

df.iat[1, 2]
    Access single value by index

df.at[4, 'A']
    Access single value by label
```

Logic in Python (and pandas)				engine="python")	regex (Regular Expressions) Examples	
<	Less than	!=	Not equal to	'\.'	Matches strings containing a period '.'	
>	Greater than	df.column.isin(values)	Group membership	'Length\$'	Matches strings ending with word 'Length'	
==	Equals	pd.isnull(obj)	Is NaN	'^Sepal'	Matches strings beginning with the word 'Sepal'	
<=	Less than or equals	pd.notnull(obj)	Is not NaN	'^x[1-5]\$'	Matches strings beginning with 'x' and ending with 1,2,3,4,5	
>=	Greater than or	&, , ~, ^, df.any(), df.all()	Logical and, or, not, xor, any, all	'^(?!Species\$).*\$'	Matches strings except the string 'Species'	

Group Data



df.groupby(by="col")
Return a GroupBy object, grouped by values in column named "col".

df.groupby(level="ind")
Return a GroupBy object, grouped by values in index level named "ind".

All of the summary functions listed above can be applied to a group. Additional GroupBy functions:
size() Size of each group.
agg(function) Aggregate group using

The examples below can also be applied to groups. In this case, the function is applied on a per-group basis, and the returned vectors are of the length of the original DataFrame.

shift(1)

Copy with values shifted by 1.

rank(method='dense')

Ranks with no gaps.

rank(method='min')

Ranks. Ties get min rank.

rank(pct=True)

Ranks rescaled to interval [0, 1].

rank(method='first')

Ranks. Ties go to first value.

shift(-1)

Copy with values lagged by 1.

cumsum()

Cumulative sum.

cummax()

Cumulative max.

cummin()

Cumulative min.

cumprod()

Summarize Data

df['w'].value_counts()

Count number of rows with each unique value of variable
len(df) # of rows in DataFrame.

df.shape

Tuple of # of rows, # of columns in DataFrame.

df['w'].nunique()

of distinct values in a column.

df.describe()

Basic descriptive and statistics for each column (or GroupBy).

df.info()

Prints a concise summary of the DataFrame.

df.memory_usage()

Prints the memory usage of each column in the DataFrame.

df.dtypes()

Prints a Series with the dtype of each column in the DataFrame.



pandas provides a large set of **summary functions** that operate on different kinds of pandas objects (DataFrame columns, Series, GroupBy, Expanding and Rolling (see below)) and produce single values for each of the groups. When applied to a DataFrame, the result is returned as a pandas Series for each column. Examples:

sum()

Sum values of each object.

count()

Count non-NA/null values of each object.

median()

Median value of each object.

quantile([0.25, 0.75])

Quantiles of each object.

apply(function)

Apply function to each object.

min()

Minimum value in each object.

max()

Maximum value in each object.

mean()

Mean value of each object.

var()

Variance of each object.

std()

Standard deviation of each object.

Handling Missing Data

df.dropna()

Drop rows with any column having NA/null data.

df.fillna(value)

Replace all NA/null data with value.

Make New Columns



df.assign(Area=lambda df: df.Length*df.Height)

Compute and append one or more new columns.

df['Volume'] = df.Length*df.Height*df.Depth

Add single column.

pd.qcut(df.col, n, labels=False)

Bin column into n buckets.



pandas provides a large set of **vector functions** that operate on all columns of a DataFrame or a single selected column (a pandas Series). These functions produce vectors of values for each of the columns, or a single Series for the individual Series. Examples:

max(axis=1)

Element-wise max.

clip(lower=-10, upper=10)

Trim values at input thresholds

min(axis=1)

Element-wise min.

abs()

Absolute value.

Windows

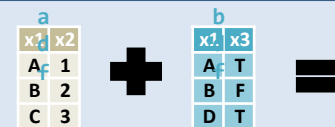
df.expanding()

Return an Expanding object allowing summary functions to be applied cumulatively.

df.rolling(n)

Return a Rolling object allowing summary functions to be applied to windows of length n.

Combine Data Sets



Standard Joins

x1	x2	x3
A	1	T
B	2	F
C	3	NaN

pd.merge(adf, bdf, how='left', on='x1')
Join matching rows from bdf to adf.

x1	x2	x3
A	1.0	T
B	2.0	F
D	NaN	T

pd.merge(adf, bdf, how='right', on='x1')
Join matching rows from adf to bdf.

x1	x2	x3
A	1	T
B	2	F

pd.merge(adf, bdf, how='inner', on='x1')
Join data. Retain only rows in both sets.

x1	x2	x3
A	1	T
B	2	F
C	3	NaN
D	NaN	T

pd.merge(adf, bdf, how='outer', on='x1')
Join data. Retain all values, all rows.

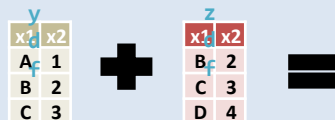
Filtering Joins

x1	x2
A	1
B	2

adf[adf.x1.isin(bdf.x1)]
All rows in adf that have a match in bdf.

x1	x2
C	3

adf[~adf.x1.isin(bdf.x1)]
All rows in adf that do not have a match in bdf.



Set-like

x1	x2
B	2
C	3

pd.merge(ydf, zdf)
Rows that appear in both ydf and zdf (Intersection).

x1	x2
A	1
B	2
C	3
D	4

pd.merge(ydf, zdf, how='outer')
Rows that appear in either or both ydf and zdf (Union).

x1	x2
A	1

pd.merge(ydf, zdf, how='outer', indicator=True)
.query('_merge == "left_only"')
.drop(columns=['_merge'])
Rows that appear in ydf but not zdf (Setdiff).

Plotting

df.plot()

Plot a line graph for the DataFrame.



df.plot.bar()

Plot a bar graph for the DataFrame.



df.plot(subplots=True)

Separate into different graphs for each column in the DataFrame.

df.plot(title="Graph of A against B")

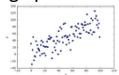
Sets the title of the graph.

df.plot(subplots=True, title=['col1', 'col2', 'col3'])

Arguments can be combined for more flexibility when graphing, this would plot a separate line graph for of column of a 3-columned DataFrame. The first string in the list of titles applies to the graph of the left-most column.

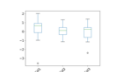
df.plot.scatter(x='w', y='h')

Plot a scatter graph of the DataFrame.



df.plot.boxplot()

Plot a scatter graph of the DataFrame.



df.plot(cumulative=True)

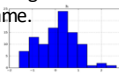
Creates a cumulative plot

df.plot(bins=30)

Set the number of bins into which data is grouped (histograms)

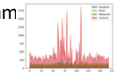
df.plot.hist()

Plot a histogram of the DataFrame.



df.plot.area()

Plot an area graph of the DataFrame.



df.plot.pie()

Plot a pie chart of the DataFrame.



df.plot.hexbin()

Plot a hexbin graph of the DataFrame.



df.plot(stacked=True)

Stacks the data for the columns on top of each other. (bar, barh and area only)

df.plot(alpha=0.5)

Sets the transparency of the plot to 50%.

Changing Type

pd.to_numeric(data)

Convert non-numeric types to numeric.

pd.to_datetime(data)

Convert non-datetime types to datetime type

pd.to_timedelta(data)

Convert non- timedelta types to timedelta

df.astype(type)

Convert data to (almost) any given type including categorical

df.infer_objects()

Attempts to infer a better type for object type data.

df.convert_dtypes()

Convert columns to best possible dtypes

Datetime

With a Series containing data of type datetime, the dt accessor is used to get various components of the datetime values:

s.dt.year

Extract the year

s.dt.month

Extract the month as an integer.

s.dt.day

Extract the day (int) from the date.

s.dt.quarter

Find which quarter the date lies in.

s.dt.hour

Extract the hour.

s.dt.minute

Extract the minute.

s.dt.second

Extract the second.

Mapping

Apply a mapping to every element in a DataFrame or Series, useful for recategorizing or transforming data.

s.map(lambda x: 2*x)

Returns a copy of the series where every entry is doubled

df.apply(lambda s: s.max() - s.min(), axis=1)

Returns a Series with the difference of the maximum and minimum values of each row of the DataFrame

Series String Operations

Similar to python string operations, except these are vectorized to apply to the entire Series efficiently.

s.str.count(pattern)

Returns a series with the integer counts in each element.

s.str.get(index)

Returns a series with the data at the given index for each element.

s.str.join(sep)

Returns a series where each element has been concatenated.

s.str.title()

Converts the first character of each word to be a capital.

s.str.len()

Returns a series with the lengths of each element.

s.str.cat()

Concatenate elements into a single string

s.str.partition(sep)

Splits the string on the first instance of the separator

s.str.slice(start, stop, step)

Slices each string

s.str.replace(pat, rep)

Use regex to replace patterns in each string.

s.str.isalnum()

Checks whether each element is alpha-numeric

Input/Output

Common file types for data input include CSV, JSON, HTML which are human-readable, while the common output types are usually more optimized for performance and scalability such as feather, parquet and HDF.

df = pd.read_csv(filepath)

Read data from csv file

df = pd.read_html(filepath)

Read data from html file

df =

pd.read_excel(filepath)

Read data from xls (and related) files

df = pd.read_sql(filepath)

Read data from sql file

pd.read_clipboard()

df.to_parquet(filepath)

Write data to parquet file

df.to_feather(filepath)

Write data to feather file

df.to_hdf(filepath)

Write data to HDF file

df.to_clipboard()

Copy object to the system clipboard

Frequently Used Options

Pandas offers some 'options' to globally control how Pandas behaves, display etc. Options can be queried and set via:

pd.options.option_name (where *option_name* is the name of an option). For example:

pd.options.display.max_rows = 20

Set the **display.max_rows** option to 20.

get_option(option)

Fetch the value of the given option.

set_option(option)

Set the value of the given option.

reset_option(options)

Reset the values of all given options to default settings.

describe_option(options)

Print descriptions of given options.

option_context(options)

Execute code with temporary option settings that revert to prior settings after execution.

Display options

display.max_rows

The maximum number of rows displayed in pretty-print.

display.max_columns

The maximum number of columns displayed in pretty-print.

display.expand_frame_repr

Controls whether the DataFrame representation stretches across pages.

display.large_repr

Controls whether a DataFrame that exceeds maximum rows/columns is truncated or summarized

display.precision

The output display precision in decimal places.

display.max_colwidth

The maximum width of columns, longer cells will be truncated.

display.max_info_columns

The maximum number of columns displayed after calling **info()**.

display.chop_threshold

Sets the rounding threshold to zero when displaying a Series/DataFrame.

display.colheader_justify

Controls how column headers are justified.

¿Matplotlib?

Matplotlib es una **librería de visualización de datos en Python**.

Permite generar gráficos de todo tipo:

- Líneas
- Barras
- Histogramas
- Diagramas de dispersión (*scatter plots*)
- Gráficos más avanzados (mapas de calor, 3D, etc.)

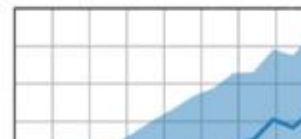
¿Por qué usar Matplotlib?

- Es muy flexible: se puede personalizar casi todo (colores, etiquetas, tamaños, estilos).
- Se integra fácilmente con Pandas y NumPy, por lo que es ideal para análisis de datos.
- Es un estándar en la comunidad científica y de datos abiertos: si compartes tu código, otros podrán reproducir tus gráficos.

matplotlib

Plot types User guide Tutorials Examples Reference Contribute Releases

<https://matplotlib.org>



Matplotlib: Visualization with Python

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

- Create [publication quality plots](#).
- Make [interactive figures](#) that can zoom, pan, update.
- Customize [visual style](#) and [layout](#).
- Export to [many file formats](#).
- Embed in [JupyterLab](#) and [Graphical User Interfaces](#).
- Use a rich array of [third-party packages](#) built on Matplotlib.

[Try Matplotlib \(on Binder\)](#)



reference



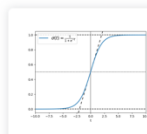
[Cheat Sheets](#)



[Documentation](#)

<https://matplotlib.org/cheatsheets/>

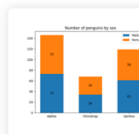
Lines, bars and markers



Infinite lines



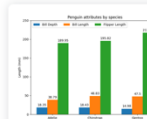
Bar chart with individual bar colors



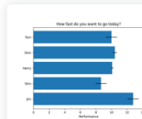
Bar chart with labels



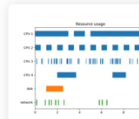
Stacked bar chart



Grouped bar chart with labels



Horizontal bar chart



Broken horizontal bars



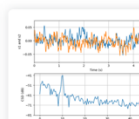
CapStyle



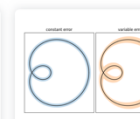
Markers



Line styles



Line colors



Line widths

```
@Article{Hunter:2007,  
  Author = {Hunter, J. D.},  
  Title = {Matplotlib: A 2D graphics environment},  
  Journal = {Computing in Science & Engineering},  
  Volume = {9},  
  Number = {3},  
  Pages = {90--95},  
  abstract = {Matplotlib is a 2D graphics package used for Python for  
    application development, interactive scripting, and publication-quality  
    image generation across user interfaces and operating systems.},  
  publisher = {IEEE COMPUTER SOC},  
  doi = {10.1109/MCSE.2007.55},  
  year = {2007}  
}
```

```
import matplotlib.pyplot as plt
```

```
# Datos de ejemplo
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [2, 4, 6, 8, 10]
```

```
# Crear un gráfico de líneas
```

```
plt.plot(x, y)
```

```
# Añadir título y etiquetas
```

```
plt.title("Ejemplo simple con Matplotlib")
```

```
plt.xlabel("Eje X")
```

```
plt.ylabel("Eje Y")
```

```
# Mostrar el gráfico
```

```
plt.show()
```

Quick start

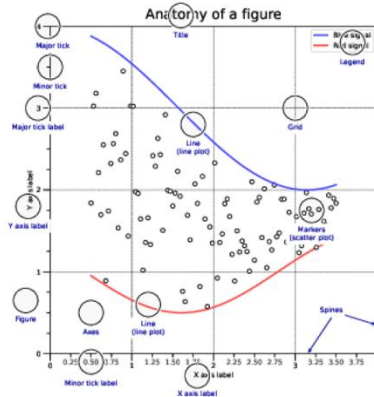
```
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
```

```
X = np.linspace(0, 2*np.pi, 100)
Y = np.cos(X)
```

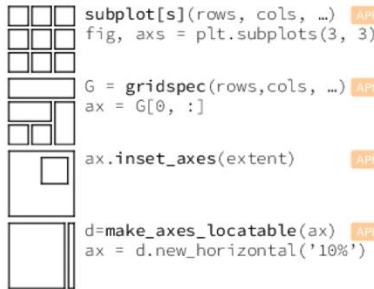
```
fig, ax = plt.subplots()
ax.plot(X, Y, color='green')
```

```
fig.savefig("figure.pdf")
plt.show()
```

Anatomy of a figure



Subplots layout



Getting help

matplotlib.org
github.com/matplotlib/matplotlib/issues
discourse.matplotlib.org
stackoverflow.com/questions/tagged/matplotlib
<https://gitter.im/matplotlib/matplotlib>
twitter.com/matplotlib
[Matplotlib users mailing list](https://matplotlib.org/users/mailling-list)

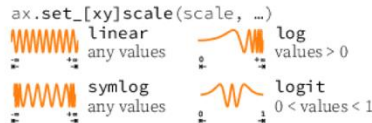
Basic plots



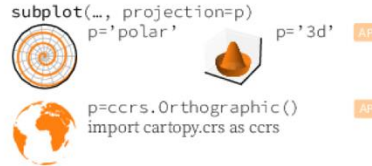
Advanced plots



Scales



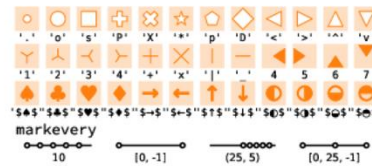
Projections



Lines



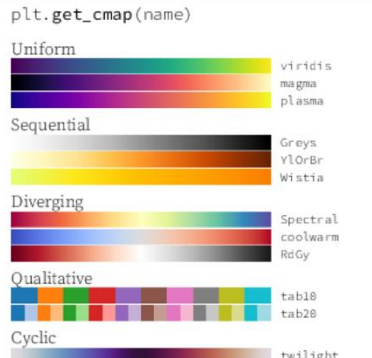
Markers



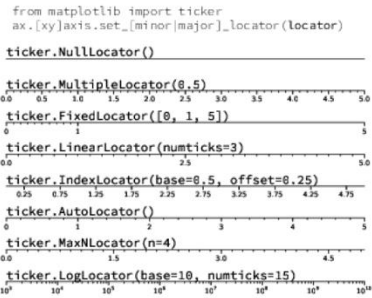
Colors



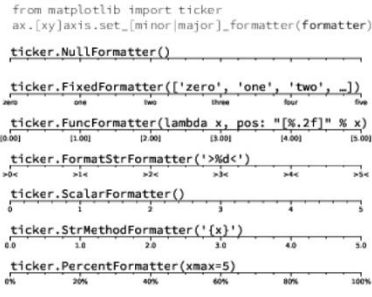
Colormaps



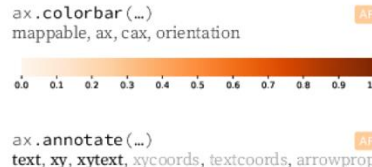
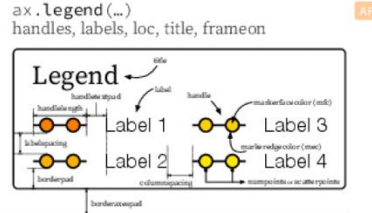
Tick locators



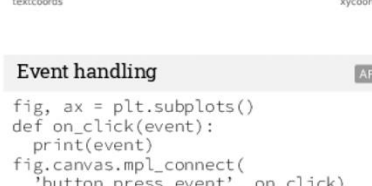
Tick formatters



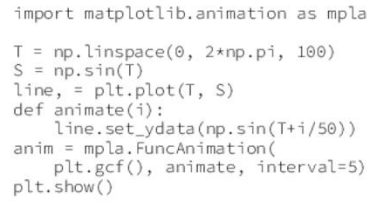
Ornaments



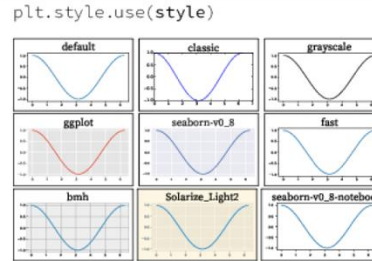
Event handling



Animation



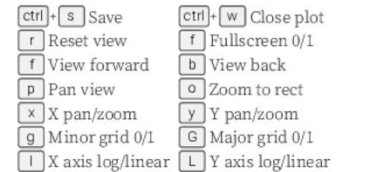
Styles



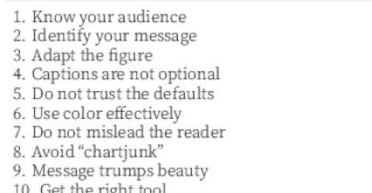
Quick reminder



Keyboard shortcuts



Ten simple rules



Abre el cuaderno de Introducción a Pandas



Abre el cuaderno de introducción a matplotlib

matplotlib

